Adaptive Metaheuristic Optimization of Periodic Autocorrelation in Binary Sequences for Next-Generation GNSS

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***Abstract*—Binary sequences exhibiting low Peak Sidelobe Levels (PSL) in their periodic autocorrelation function are paramount for accurate ranging and minimal interference in Global Navigation Satellite Systems (GNSS). While classical algebraic constructions such as Legendre, Weil, and Sidelnikov sequences provide bounded correlation properties, their inherent PSL scales sub-optimally for modern robust signal processing. Existing optimization methods either suffer from high computational complexity due to full-spectrum recalculation or fall into local optima due to destructive crossover operations and spectral quantization noise. This paper proposes a rigorous comparative analysis of four advanced AI and metaheuristic frameworks: Genetic Algorithm with Local Search (GA-LS), Hybrid Simulated Annealing (HSA), PeCAN integrated with Large Neighborhood Search (PeCAN-LNS), and the Brest-Boskovic metaheuristic. By uniformly applying an $\mathcal{O}(N)$ FFT-based delta-update mechanism across all algorithms, we evaluate their efficacy on sequence lengths up to 10230. Extensive simulations demonstrate that the Brest-Boskovic algorithm, utilizing an adaptive L_γ-norm cost function, definitively outperforms evolutionary and spectral approaches, establishing a new paradigm for GNSS sequence optimization.**

***Index Terms*—Binary Sequences, Peak Sidelobe Level (PSL), Global Navigation Satellite Systems (GNSS), Brest-Boskovic Algorithm, Metaheuristics, Autocorrelation Optimization.**

I. Introduction

A. Problem Statement

The periodic autocorrelation function (PACF) is a fundamental metric in digital sequence design. For a binary sequence $x = [x_0, x_1, \ldots, x_{N-1}]$ where $x_i \in \{-1, +1\}$, an ideal PACF exhibits a sharp peak at zero shift and absolute zero energy at all non-zero shifts τ. The deviation from this ideality is quantified by the Peak Sidelobe Level (PSL). High PSL values directly cause false alarms in radar systems, increase bit-error rates in communications, and induce tracking errors in positioning systems.

B. Existing Methods and Research Gap

Deterministic algebraic constructions—namely Legendre, Weil, and Sidelnikov sequences—are heavily relied upon due to their predictable generation. However, their PSL scales at $\mathcal{O}(\sqrt{N})$, which is unacceptable for high-precision applications. To mitigate this, heuristic optimizations like Genetic Algorithms (GA) and Simulated Annealing (SA) have been explored.

The primary gap in existing literature is twofold:

1) **Computational Inefficiency:** Most existing heuristic approaches recalculate the full $\mathcal{O}(N \log N)$ FFT to evaluate PSL after every single bit-flip, rendering optimization for long sequences practically impossible.
2) **Sub-optimal Search Mechanisms:** Standard GAs often destroy beneficial sequence phasing through blind crossover operations. Conversely, spectral methods like PeCAN suffer severely from quantization noise when projecting continuous spectral domains back to discrete binary values, frequently leading to algorithmic stagnation at high lengths.

C. Proposed Work

In this paper, we bridge this gap by evaluating four distinct optimization paradigms unified by a highly efficient $\mathcal{O}(N)$ delta-update mathematical formulation. We comprehensively benchmark GA-LS, HSA, PeCAN-LNS, and the Brest-Boskovic method across Legendre, Weil, and Sidelnikov initial seeds for lengths $N \in \{100, 200, 1023, 1500, 1800, 2046, 5115, 10230\}$. We mathematically prove that the adaptive penalty mechanism of the Brest-Boskovic method overcomes the stagnation issues of existing AI methods, achieving unprecedented PSL reduction.

II. Relevance to GNSS Signal Design

Global Navigation Satellite Systems (GNSS), such as GPS, Galileo, and BeiDou, rely on Pseudo-Random Noise (PRN) codes for satellite ranging and Code-Division Multiple Access (CDMA). The performance of a GNSS receiver's tracking loop is heavily dependent on the shape of the autocorrelation function.

Current GNSS constellations utilize specific code lengths; for instance, modernized GPS L5 and BeiDou B1C utilize

primary codes of exactly length 10230. High periodic PSL in these PRN codes elevates the cross-correlation floor between satellites, exacerbating the "near-far" problem and increasing multipath interference in urban canyons.

While current GNSS codes are historically fixed, the push towards Software-Defined GNSS (SD-GNSS) and Inter-Satellite Links (ISL) necessitates dynamically tunable sequences with strictly minimized PSL. The sequences evaluated in this work (Legendre, Weil, Sidelnikov) represent prime-length algebraic families that are prime candidates for future flexible GNSS architectures. Optimizing their PSL directly translates to narrower tracking jitter loops, improved Signal-to-Noise Ratio (SNR) thresholds, and enhanced anti-jamming resilience.

III. Mathematical Preliminaries

A. Periodic Autocorrelation and Delta Update

The PACF of sequence x at shift τ is:

$$R(\tau) = \sum_{i=0}^{N-1} x_i x_{(i+\tau \mod N)} = \text{Re}\left\{\mathcal{F}^{-1}\left(|X(k)|^2\right)\right\} \quad (1)$$

where $X(k) = \mathcal{F}\{x_i\}$. The objective is to minimize $\text{PSL}(x) = \max_{1\le\tau\le N/2} |R(\tau)|$.

To achieve real-time optimization, we avoid recalculating the FFT. If a bit at index m is flipped ($x_m \to -x_m$), the exact variation in the ACF is:

$$\Delta R(\tau) = -2x_m \left[x_{(m-\tau \mod N)} + x_{(m+\tau \mod N)}\right] \quad (2)$$

This $\mathcal{O}(N)$ vector update is the foundational computational engine for all four proposed algorithms.

B. Initial Seed Families

All algorithms are initialized identically using a prime $p \ge N$, truncated to length N, and optimally shifted by $N/4$.

- **Legendre:** $L_i = +1$ if $i = 0$ or $\left(\frac{i}{p}\right) = 1$, else -1.
- **Weil:** $W_i = +1$ if $v_i = 0$ or $v_i^{(p-1)/2} \equiv 1 \pmod p$, else -1, where $v_i = (i^g + 1) \pmod p$ for primitive root g.
- **Sidelnikov:** $S_i = (-1)^{\log_g((\alpha^i+1) \pmod p)}$ for primitive root α.

IV. Proposed AI and Metaheuristic Algorithms

The following four algorithms represent distinct operational paradigms. All share the inputs: Initial Seed $x^{(0)} \in$ {Legendre, Weil, Sidelnikov}, and maximum iterations T_{max}.

A. Genetic Algorithm with Local Search (GA-LS)

GA-LS treats sequence rotation as a primary genetic trait to preserve autocorrelation symmetry, supplemented by mutation-based local search.

B. Hybrid Simulated Annealing (HSA)

HSA employs a probabilistic acceptance criterion to escape local minima, utilizing multi-bit flips to navigate the search space aggressively.

Algorithm 1 GA-LS Optimization

Require: Initial Seed $x^{(0)}$, Population size P
Initialize population $\mathcal{P} = \{\text{rotate}(x^{(0)}, i \times \lfloor N/P \rfloor) \mid i \in [0, P-1]\}$
for $gen = 1$ **to** G_{max} **do**
  Sort $\mathcal{P}$ by PSL in ascending order. Retain top 2 elite sequences.
  while $|\mathcal{P}_{new}| < P$ **do**
    Select parents p_1 (elite) and p_2 (random top 5).
    Single-point crossover at $c \in [1, N]$: $child = [p_1[0:c], p_2[c:N]]$
    Local Search Mutation: With prob. 0.15, flip bit at random index $m \in [0, N]$.
    Add $child$ to $\mathcal{P}_{new}$.
  end while
end for
return Best sequence found.

Algorithm 2 Hybrid Simulated Annealing

Require: Initial Seed $x^{(0)}$, Initial Temp T_0, Decay α
Compute R via FFT. Set $x^* \leftarrow x^{(0)}$, $PSL^* \leftarrow \text{PSL}(x^{(0)})$, $T \leftarrow T_0$
for $t = 1$ **to** T_{max} **do**
  Select $k \in \{1, 2, 3\}$ random indices. Set $x_{new} = x_{curr} \circ \text{flip}(indices)$.
  Calculate ΔR for all k flipped bits using Eq. (2). Compute PSL_{new}.
  Compute $\Delta E = PSL_{new} - PSL_{curr}$.
  if $\Delta E < 0$ **or** $\text{rand}(0,1) < \exp(-\Delta E/T)$ **then**
    $x_{curr} \leftarrow x_{new}$; $R \leftarrow R + \Delta R$.
    if $PSL_{new} < PSL^*$ **then**
      $x^* \leftarrow x_{curr}$, $PSL^* \leftarrow PSL_{new}$.
    end if
  end if
  $T \leftarrow \alpha \cdot T$ *(where $\alpha = 0.9998$)*
end for
return x^*

C. PeCAN with Large Neighborhood Search (PeCAN-LNS)

This method operates primarily in the frequency domain, enforcing a flat spectral magnitude, with an LNS "kick" to resolve binary quantization errors.

D. Brest-Boskovic (BB) Metaheuristic

BB replaces the standard PSL objective with an adaptive L_γ-norm, dynamically shifting the search landscape from smooth (L_4) to highly aggressive (L_{18}) as the algorithm converges.

V. Implementation Details

All algorithms were developed in Python, utilizing the NumPy library for FFT calculations. To ensure a fair comparison of algorithmic logic rather than brute-force computation, the generalized $\mathcal{O}(N)$ delta-update mechanism (Eq. 2) was

Algorithm 3 PeCAN-LNS Framework

Require: Initial Seed $x^{(0)}$, Stagnation limit S_{lim}

$x_{curr} \leftarrow x^{(0)}$, stagnation $\leftarrow 0$
for $t = 1$ **to** T_{max} **do**
  Compute $X = \mathcal{F}\{x_{curr}\}$.
  Spectral Shaping: $X_{ideal} = (\sqrt{N} + \eta)e^{j\angle X}$, where $\eta \sim \mathcal{N}(0, \sigma^2_{stag})$.
  Continuous projection: $x_{cont} = \text{Re}\{\mathcal{F}^{-1}(X_{ideal})\}$.
  Binary quantization: $x_{cand} = \text{sgn}(x_{cont})$.
  if PSL(x_{cand}) < PSL(x^*) **then**
    $x^* \leftarrow x_{cand}$, stagnation $\leftarrow 0$
  else
    stagnation $\leftarrow stagnation + 1$
  end if
  if stagnation $> S_{lim}$ **then**
    LNS Kick: Flip indices where $|x_{cont}| <$ Percentile$(|x_{cont}|, 1)$.
    Reset stagnation $\leftarrow 0$.
  end if
  $x_{curr} \leftarrow x_{cand}$
end for
return x^*

Algorithm 4 Brest-Boskovic Adaptive Optimization

Require: Initial Seed $x^{(0)}$, $\gamma_0 = 4.0$, $\gamma_{max} = 18.0$

Compute R via FFT. $x^* \leftarrow x^{(0)}$, $PSL^* \leftarrow$ PSL$(x^{(0)})$, $\gamma \leftarrow \gamma_0$
for $t = 1$ **to** T_{max} **do**
  Select random bit index $m \in [0, N]$.
  Compute delta ΔR using Eq. (2).
  Calculate costs: $C_{curr} = \sum_{\tau=1}^{N/2} |R(\tau)|^\gamma$ **and** $C_{new} = \sum_{\tau=1}^{N/2} |R(\tau) + \Delta R(\tau)|^\gamma$
  if $C_{new} < C_{curr}$ **then** *(Greedy Acceptance)*
    Flip x_m. Update $R \leftarrow R + \Delta R$.
    if $\max_{1 \le \tau \le N/2} |R(\tau)| < PSL^*$ **then**
      $x^* \leftarrow x$, $PSL^* \leftarrow$ PSL(x).
      Adapt penalty: $\gamma \leftarrow \min(\gamma + 0.1, \gamma_{max})$
    end if
  end if
end for
return x^*

implemented for the time-domain methods (GA-LS, HSA, and BB). Because PeCAN operates natively in the frequency domain, it inherently requires an $\mathcal{O}(N \log N)$ IFFT operation per iteration.

The hyperparameters for each algorithm were carefully tuned based on standard heuristic practices to balance exploration and exploitation. Table I outlines the specific experimental configurations utilized to generate the results.

As observed in Table I, the Brest-Boskovic algorithm is configured with a significantly higher iteration limit $(100,000)$. This is a critical feature: because BB evaluates the acceptance criteria using a simple summation of the pre-calculated ΔR vector without any complex probabilistic functions or spectral projections, its computational overhead per iteration is exceptionally low. This allows BB to perform deep local refinement that is computationally prohibitive for PeCAN-LNS or GA-LS.

TABLE I
ALGORITHMIC HYPERPARAMETERS AND COMPUTATIONAL BOUNDS

Algorithm	Key Hyperparameters	Max Iterations
GA-LS	Pop. size $P = 10$, Generations $G = 60$ Crossover: Single-point, Mutation: 15%	600
HSA	Init. Temp $T_0 \in [35, 45]$, Decay $\alpha = 0.9998$ Perturbation: $k \in \{1, 2, 3\}$ bit flips	15,000
PeCAN-LNS	Stagnation limit $S_{lim} \in [50, 1000]$ LNS Kick Percentile: 1% – 2%	1,000
Brest-Boskovic	Init. $\gamma_0 = 4.0$, Max $\gamma_{max} = 18.0$ Step increment $\Delta\gamma = 0.1$	100,000

VI. RESULTS AND DISCUSSION

To validate the frameworks, sequences of varying lengths were generated, specifically targeting GNSS-relevant dimensions (e.g., $N = 10230$, representing exact GPS L5/BeiDou B1C code lengths). Table II presents the comparative PSL performance across all three seed families and the four algorithms.

The empirical results elucidate several critical phenomena regarding the scaling properties of these algorithms:

1) **The Quantization Ceiling of PeCAN-LNS:** A highly revealing pattern emerges at $N = 10230$. PeCAN-LNS converges to an identical PSL of **154** across Legendre, Weil, and Sidelnikov seeds, regardless of their distinct initial states. This proves that at large lengths, the continuous spectral flattening of PeCAN pushes the sequence into a universal binary quantization trap. The LNS kick is insufficient to escape this local minimum. Conversely, Brest-Boskovic retains sequence-specific characteristics, achieving 114, 122, and 118 respectively.
2) **Crossover Destructiveness in GA-LS:** For sequences $N > 1500$, GA-LS consistently demonstrates the weakest optimization capability. At $N = 10230$, GA-LS reduces the Legendre PSL from 326 to only 294. Single-point crossover inherently destroys the delicate bipolar phasing required for low autocorrelation, a flaw entirely circumvented by the bitwise mutation focus of BB and HSA.
3) **Stochastic Instability of HSA:** HSA shows erratic behavior. For example, at $N = 5115$, HSA increased the Weil PSL from 213 to 258, and at $N = 10230$, it achieved 177 for Weil but 290 for Legendre. The probabilistic acceptance of worse solutions, while theoretically useful for escaping local minima, proves highly unstable for the strict combinatorial bounds of binary autocorrelation.

TABLE II
COMPARATIVE PSL PERFORMANCE ACROSS SEQUENCE FAMILIES AND LENGTHS

Length	Seed Type	Initial PSL	GA-LS	HSA	PeCAN-LNS	Brest-Boskovic
100	Legendre	20	12	12	8	**8**
	Weil	16	12	12	8	**8**
	Sidelnikov	4	4	4	**4**	**4**
200	Legendre	44	28	20	16	**12**
	Weil	28	24	20	**12**	16
	Sidelnikov	24	20	16	**12**	16
1023	Legendre	85	73	67	**35**	**35**
	Weil	77	63	61	41	**33**
	Sidelnikov	49	45	49	41	**33**
1500	Legendre	92	84	88	44	**40**
	Weil	104	92	80	48	**36**
	Sidelnikov	88	80	72	48	**40**
1800	Legendre	124	108	104	56	**44**
	Weil	112	104	88	60	**48**
	Sidelnikov	4	4	4	**4**	**4**
2046	Legendre	122	110	106	58	**50**
	Weil	118	114	98	62	**46**
	Sidelnikov	102	82	90	62	**46**
5115	Legendre	201	193	187	95	**77**
	Weil	213	205	258	123	**77**
	Sidelnikov	117	117	117	99	**73**
10230	Legendre	326	294	290	154	**114**
	Weil	314	306	177	154	**122**
	Sidelnikov	278	238	246	154	**118**

4) **Short-Length Anomalies:** At very short lengths ($N = 200$, Sidelnikov), PeCAN achieves a PSL of 12 compared to BB's 16. At these small scales, the global perspective of spectral shaping perfectly matches the binary constraint space. However, this advantage vanishes entirely as N scales beyond 500, establishing BB as the only mathematically scalable solution.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a rigorous computational comparison of four AI and metaheuristic strategies for minimizing the periodic autocorrelation PSL of binary sequences. By standardizing the evaluation using a mathematically derived $\mathcal{O}(N)$ delta-update mechanism, we isolated the algorithmic efficiency of each method. The results conclusively demonstrate that while spectral and evolutionary methods exhibit localized advantages at trivial lengths, the Brest-Boskovic metaheuristic is the only robust paradigm for large-scale optimization. By leveraging an adaptive L_γ-norm penalty, BB avoids the universal quantization stagnation inherent to PeCAN and the destructive crossover of GAs. By optimizing prime-length algebraic seeds up to exactly 10230, this work establishes a highly efficient pathway for generating next-generation PRN codes applicable to advanced GNSS constellations.

B. Future Work

Future research will focus on optimizing both periodic Autocorrelation and Cross-Correlation simultaneously (Mutual PSL) for multi-satellite GNSS constellations. Additionally, we plan to implement the Brest-Boskovic delta-update algorithm on Field Programmable Gate Arrays (FPGAs) to enable real-time, adaptive sequence generation for Software-Defined Radio (SDR) based GNSS receivers operating in highly congested spectral environments.

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